

BLMIM Phase 2D — Seldon Factor Model: Seven Mathematical Frameworks for Civilisational-Variable Equity Analysis

BLMIM Phase 2D · Seldon Factor Model · 7 Frameworks · EDA Sector Live
Analysis · BL3 MySQL · CivilisationOS

PhD Due Diligence Report | BLRI-BLMIM-PHASE2D-AUG2026

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Six-Method Framework: Quantitative · Qualitative · Ethnographic · Superforecasting · Adversarial-Collaborative · Seldon Thesis

*RAG Corpus: EDGAR 139,756ch · WSJ 336,316ch · NIKKEI 397,454ch · FT 110,100ch · ST_PHD 61,295ch | Sources:
FinancialMathConvergence · GeometryOfEverything · MathCivilisation · CivilisationCodex*

PART I — SELDON THESIS: THE MATHEMATICAL ARCHITECTURE OF EQUITY ANALYSIS

BLMIM Phase 2D implements the Seldon Factor Model — the formal quantitative bridge between the CivilisationOS Phi(C) civilisational health framework and individual equity valuation. The thesis: if the seven Seldon variables (A, K, I, L, ξ , Ω , E) govern the long-run health trajectory of civilisation, then equities whose fundamental operations are structurally aligned with rising K, rising I, and declining Ω must systematically outperform the market over any horizon in which the civilisational trajectory is not in free-fall collapse.

The mathematical foundations of Phase 2D are drawn from four PhD theses ingested into ST_PHD_RAG during August 2026: BlueLotus_FinancialMathConvergence_PhD_Aug2026 (convergence attractor theory, contracted backlog as revenue stabiliser), BlueLotus_GeometryOfEverything_PhD_Aug2026 (geometric phase space trajectory classification), BlueLotus_MathCivilisation_PhD_Aug2026 (K variable as primary long-duration equity driver), and BlueLotus_CivilisationCodex_PhDThesis_Aug2026 (formal 7-variable Phi(C) codification and sector factor loading matrix).

CURRENT PHI(C) STATE (from blmim_seldon_state_20260815.json, Phase 2C extraction): $A=0.2528 \cdot K=0.84 \cdot I=0.525 \cdot L=0.5502 \cdot \xi=0.60 \cdot \Omega=0.3839 \cdot E=0.54 \rightarrow \text{Phi(C)}=0.3728$. PHI_PLATEAU regime

($|d\Phi|=0.0028 < \text{threshold } 0.005$). In PHI_PLATEAU, the macro environment is neutral — K-factor selection dominates. $K=1.00$ Enabler equities are the primary long-duration structural overweight.

PHI(C) FUNCTION AND VARIABLE DEFINITIONS

Variable	Name	2026 Value	Status	Equity Implication
A	Asabiyyah / Social Cohesion	0.2528	⚠️ WARN — below 0.50 floor	Consumer Discretionary impaired · Financial_Banks at risk · Defence unaffected
K	Knowledge Capital	0.84	HIGH — trending toward 1.00	EDA_Software + Semiconductor primary beneficiaries · $K=1.00$ Enablers premium
I	Institutional Resilience	0.525	OK — above 0.40 floor	Financial_Banks + EDA_Software benefit from rule-of-law premium
L	Legitimacy	0.5502	OK — above 0.45 floor	Consumer sectors benefit; declining trend reduces L-loaded sectors
ξ	Gaia Coefficient	0.60	ADEQUATE — above 0.35 floor	Energy transition + nuclear = ξ uplift; fossil extraction = ξ penalty
Ω	Existential Risk	0.3839	WARN — approaching 0.60 override	Defence_Aerospace benefits directly (+0.80 Ω -loading) · $\Omega < 0.60$ = no override
E	Elite Overproduction	0.54	WARN — approaching 0.70 ceiling	High E suppresses Consumer_Discretionary and Real_Estate valuations

PART II — THE SEVEN MATHEMATICAL FRAMEWORKS

Framework 1: Hurst Exponent (R/S Analysis) — Convergence Attractor Proximity

Mathematical basis: FinancialMathConvergence §3.2. The Hurst exponent measures whether a price series is mean-reverting (near a convergence attractor) or trending (far from it). Computed via Rescaled Range analysis: $R/S(n) = C \times n^H$, where $\log(R/S) = \log(C) + H \times \log(n)$.

H Range	Interpretation	Convergence Score	Investment Signal
$H < 0.45$	Anti-persistent — mean-reverting; near convergence attractor	$1.0 - H/0.45$	HURST_MEAN_REVERSION +0.5 added to composite
$H \in [0.45, 0.55]$	Geometric Brownian Motion — no dominant attractor	0.50	Neutral — no signal generated
$H > 0.55$	Persistent — trending; momentum dominant; far from attractor	$\max(0, 1 - (H-0.5)/0.5)$	Momentum continuation — watch for trend exhaustion

Limitations: R/S analysis requires 1,000+ observations for reliable $H < 0.45$ detection. For sub-2-year daily windows (500 bars), the OU process fitting is the more reliable attractor detector. Hurst is supplementary.

Framework 2: Ornstein-Uhlenbeck Process — Mean Reversion Speed

Mathematical basis: FinancialMathConvergence §4.1, GeometryOfEverything §2.3. The OU SDE: $dX = \theta(\mu - X)dt + \sigma dW$, fitted via AR(1) regression on log-prices. θ = mean reversion speed (annualised) · μ = long-run price attractor · Half-life = $\ln(2)/\theta$. Strong attractor criterion: half_life < 1 year.

The contracted backlog (FinancialMathConvergence §4.3) is the primary mechanism that creates a financial OU attractor: actual revenue cannot deviate far from Backlog/n_years without contract renegotiation. This suppresses volatility and generates a systematic P/E premium over non-backlog peers.

Framework 3: K-Intensity Factor — Knowledge Capital Equity Premium

Mathematical basis: MathCivilisation §7.4, CivilisationCodex K-variable. $K_score = 0.35 \cdot \text{norm}(R\&D/Rev) + 0.25 \cdot \log_norm(\text{Backlog}/Rev) + 0.20 \cdot \text{Subscription_pct} + 0.10 \cdot \text{Patent_intensity} + 0.10 \cdot \text{STEM_fraction}$. Weight calibration: R&D is the primary K-capital generator (0.35). Backlog preserves K and creates convergence attractor (0.25). Subscription revenue is recurring K delivery with minimal new K cost (0.20).

K-Score	Classification	Structural Interpretation	P/E Premium
≥ 0.80	K=1.00 Enabler	PrimeTime STA / Calibre DRC / ASML EUV class. Mandatory infrastructure for K variable itself. Cannot be substituted.	2–3× sector PE structural premium over 10-year horizon
≥ 0.60	K-Intensive	Significant K-capital generator. Strong R&D + backlog predictability.	+0.5–1.0× sector PE K-premium warranted
≥ 0.35	K-Standard	Moderate R&D intensity. Full cyclical exposure.	No structural K-premium; cycle-managed positioning
< 0.35	K-Minimal	Low R&D, commoditised tech. E-variable risk (graduate oversupply vs K output).	Discount to sector PE; cyclical trading only

Framework 4: Seldon Macro Factor Model — Sector Rotation Signal

Mathematical basis: CivilisationCodex §9, FinancialMathConvergence §6.2. $\text{Factor_return}(\text{sector}) = \sum_i [\text{loading}_i \times \phi_variable_i] + \phi_delta \cdot \text{momentum_adj}$. Regime logic: $\phi_delta > 0 \rightarrow$ growth rotation (K/I/A rising); $\phi_delta < 0 \rightarrow$ defensive rotation (Ω rising, A declining); $\Phi < 0.30 \rightarrow$ crisis floor (all growth signals halved); $\Omega > 0.60 \rightarrow \Omega$ -override (defensive regardless of ϕ_delta).

SELDON FACTOR LOADING MATRIX — 12 SECTORS × 7 VARIABLES

Sector	A	K	I	L	ξ	Ω	E
Technology	+0.30	+0.95	+0.50	+0.20	+0.15	-0.40	-0.10
Semiconductor	+0.25	+1.00	+0.60	+0.15	+0.20	-0.35	-0.05
EDA_Software	+0.20	+1.00	+0.70	+0.10	+0.10	-0.20	-0.02
Defence_Aerospace	-0.20	+0.40	+0.30	-0.15	+0.05	+0.80	+0.10
Healthcare_Biotech	+0.40	+0.85	+0.40	+0.30	+0.35	-0.10	-0.20

Consumer_Discretionary	+0.80	+0.20	+0.25	+0.60	+0.10	-0.70	-0.40
Consumer_Staples	+0.50	+0.10	+0.20	+0.40	+0.30	-0.40	-0.20
Financial_Banks	+0.70	+0.30	+0.85	+0.50	+0.05	-0.65	-0.30
Energy_Oil_Gas	+0.20	+0.15	+0.25	+0.20	-0.50	+0.20	+0.05
Utilities_Nuclear	+0.30	+0.35	+0.45	+0.35	+0.40	-0.20	+0.00
Industrials	+0.45	+0.40	+0.35	+0.35	+0.20	-0.30	-0.10
Real_Estate	+0.60	+0.05	+0.40	+0.55	+0.10	-0.55	-0.35

Framework 5: Geometric Phase Space Analysis — Trajectory Geometry

Mathematical basis: GeometryOfEverything §5.1–5.3 (Geometric Trajectory Analysis). Phase space coordinates: r_t = rolling mean log-return, σ_t = rolling volatility, m_t = momentum = r_t/σ_t (Sharpe-like). Trajectory curvature κ is the discrete Frenet-Serret approximation. High κ + declining σ = spiral-converging (attractor approach). High κ + rising σ = chaotic (Ω -event precursor).

Geometry	Curvature	Volatility Trend	Ω Signal	Interpretation
spiral-converging	High $\kappa > 0.5$	σ declining	0.10 (low)	Approaching convergence attractor. Contrarian entry signal.
linear-trending	Low $\kappa < 0.3$	r rising	0.20 (low)	GBM dominant. Momentum signal valid. Monitor Hurst for exhaustion.
chaotic	High $\kappa > 0.5$	σ rising > 0.005	0.75 (HIGH)	Ω -event geometric signature. Reduce position. Tail risk elevated.
stable	Low-medium κ	r, σ stable	0.15 (low)	Phase-space equilibrium. Convergence attractor engaged. K-thesis positioning.

Framework 6: Contracted Backlog Stability Score

Mathematical basis: FinancialMathConvergence §4.3. $\text{Stability_Score} = 0.70 \cdot \min(1, \text{Backlog}/\text{Revenue}/2) + 0.30 \cdot \text{Subscription_pct}$. Calibration: Synopsys FY2024 ($\$8.7\text{B} / \$6.13\text{B} = 1.42\times$) $\rightarrow 0.7517$ (Convergence-Stabilized). Implied Sharpe premium: +0.08 per 0.10 stability point (cross-section from FinancialMathConvergence convergence-premium dataset). Threshold ≥ 0.70 = Convergence-Stabilized; ≥ 0.40 = Partially-Stabilized; < 0.40 = Cyclical.

Framework 7: Phi-Delta Regime Classifier

Mathematical basis: CivilisationCodex §10. $d\Phi/dt = \phi_{\text{current}} - \phi_{\text{prior}}$; $d^2\Phi/dt^2 = \text{convexity}$. Five regimes: ACCELERATING_IMPROVEMENT (max growth) · DECELERATING_IMPROVEMENT (quality tilt) · PHI_PLATEAU (K-factor selects) · DECELERATING_DETERIORATION (near trough; accumulate $K=1.00$ Enablers) · ACCELERATING_DETERIORATION (full defensive). Priority overrides: $\Phi < 0.30$ = CRISIS_ATTRACTOR; $\Omega > 0.60$ = OMEGA_OVERRIDE ($K=1.00$ Enablers retain premium regardless).

PART III — EDA SECTOR LIVE ANALYSIS: BL3 MYSQL PRICE DATA

All price data from bluelotus3.historical_prices (BL3 MySQL). Financial ratios from EDGAR-cited annual reports ingested in ST_PHD_RAG. Analysis date: 21 August 2026. Phi(C) state: 0.3728 (PHI_PLATEAU). Phi regime: NEUTRAL_K_FACTOR_SELECT — K=1.00 Enabler equities are the primary structural overweight in the current macro environment.

K-INTENSITY & BACKLOG STABILITY — EDA SECTOR

Ticker (Company)	Sector	K-Score	K-Class	Backlog×	Stability	Classification
SNPS (Synopsys Inc.)	EDA_Software	0.8380	K=1.00 Enabler	1.42×	0.7517	Convergence-Stabilized
CDNS (Cadence Design Systems)	EDA_Software	0.8094	K=1.00 Enabler	0.89×	0.5800	Partially-Stabilized
ASML (ASML Holding N.V.)	Semiconductor	0.6550	K-Intensive	1.46×	0.6915	Partially-Stabilized
AMAT (Applied Materials Inc.)	Semiconductor	0.4561	K-Standard	0.07×	0.2207	Cyclical
KLAC (KLA Corporation)	Semiconductor	0.4380	K-Standard	0.11×	0.2189	Cyclical
LRCX (Lam Research Corporation)	Semiconductor	0.4290	K-Standard	0.09×	0.1955	Cyclical

CONVERGENCE DYNAMICS — HURST, OU PROCESS, GEOMETRIC PHASE

Ticker	N bars	Latest	Hurst H	Interpretation	OU θ	Half-life	Attractor	Phase Geometry
SNPS	224	\$413.22	0.6620	trending (far from convergen	10.83	0.064 yr	True	stable
CDNS	224	\$323.47	0.5439	random walk (no convergence	8.73	0.079 yr	True	linear-trending
ASML	166	\$1889.99	0.5633	trending (far from convergen	8.94	0.077 yr	True	stable
AMAT	166	\$536.03	0.5532	trending (far from convergen	5.48	0.126 yr	True	spiral-converging
KLAC	224	\$205.76	0.5575	trending (far from convergen	3.69	0.188 yr	True	stable
LRCX	224	\$343.84	0.5455	random walk (no convergence	3.17	0.218 yr	True	spiral-converging

SELDON MACRO FACTOR & COMPOSITE SIGNAL

Ticker	Sector	Rotation	Q-Signal	Phi Regime	Composite	Signal
SNPS	EDA_Software	+0.4341	0.15	PHI_PLATEAU	0.5057	▲ POSITIVE
CDNS	EDA_Software	+0.4341	0.20	PHI_PLATEAU	0.5585	▲▲ STRONG

ASML	Semiconductor	+0.4254	0.15	PHI_PLATEAU	0.3564	▲ POSITIVE
AMAT	Semiconductor	+0.4254	0.10	PHI_PLATEAU	0.4063	▲ POSITIVE
KLAC	Semiconductor	+0.4254	0.15	PHI_PLATEAU	0.2127	– NEUTRAL
LRCX	Semiconductor	+0.4254	0.10	PHI_PLATEAU	0.4063	▲ POSITIVE

VALUATION FROM BL3 MYSQL (ticker_fundamentals)

Ticker	Company	Latest Price	PE TTM	Mkt Cap	Sharpe+	K-Class	Snap Date
SNPS	Synopsys Inc.	\$413.22	91.1×	\$76.2B	+0.6014	K=1.00 Enabler	2026-08-21
CDNS	Cadence Design Systems	\$323.47	62.3×	\$86.4B	+0.4640	K=1.00 Enabler	2026-08-21
ASML	ASML Holding N.V.	\$1889.99	54.4×	\$672.3B	+0.5532	K-Intensive	2026-08-21
AMAT	Applied Materials Inc.	\$536.03	42.8×	\$393.8B	+0.1766	K-Standard	2026-08-21
KLAC	KLA Corporation	\$205.76	50.8×	\$242.8B	+0.1751	K-Standard	2026-08-21
LRCX	Lam Research Corporation	\$343.84	53.9×	\$388.6B	+0.1564	K-Standard	2026-08-21

SECTOR RANK BY COMPOSITE SIGNAL — SELDON SYNTHESIS

Rank	Ticker	Company	Composite	K-Score	Backlog×	Stability	Seldon Verdict
#1	CDNS	Cadence Design Systems	0.5585	0.8094	0.89×	0.5800	K=1.00 ENABLER — LONG-DURATION STRUCTURAL HOLD
#2	SNPS	Synopsys Inc.	0.5057	0.8380	1.42×	0.7517	K=1.00 ENABLER + CONVERGENCE-STABILIZED — LONG-DURATION CORE HOLD
#3	AMAT	Applied Materials Inc.	0.4063	0.4561	0.07×	0.2207	K-STANDARD — TACTICAL SIGNALS ONLY; NO STRUCTURAL K-PREMIUM
#4	LRCX	Lam Research Corporation	0.4063	0.4290	0.09×	0.1955	K-STANDARD — TACTICAL SIGNALS ONLY; NO STRUCTURAL K-PREMIUM
#5	ASML	ASML Holding N.V.	0.3564	0.6550	1.46×	0.6915	K-INTENSIVE — SECTOR OVERWEIGHT (below K=1.00 class)

#6	KLAC	KLA Corporation	0.2127	0.4380	0.11×	0.2189	K-STANDARD — TACTICAL SIGNALS ONLY; NO STRUCTURAL K-PREMIUM
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PART IV — BLMIM INTEGRATION & API GUIDE

MODULE IMPORT AND USAGE

Module path: C:\civilisationOS\scripts\blmim_financial_math.py

Import pattern: `import sys; sys.path.insert(0, r'C:\civilisationOS\scripts');` `from blmim_financial_math import full_seldon_analysis, hurst_exponent, ou_process_fit, k_intensity_score, seldon_macro_factor, geometric_phase_classifier, backlog_stability_score, phi_delta_regime, SELDON_FACTOR_LOADINGS`

Primary entry point: `full_seldon_analysis(ticker, prices, rd_revenue_ratio, backlog_revenue_usd, annual_revenue_usd, subscription_pct, sector, patent_intensity, stem_headcount_fraction)` → dict. All price arrays must come from BL3 MySQL `historical_prices` (ktype='K_DAY'). Financial ratios must come from EDGAR-cited documents in ST_PHD_RAG. ZERO TRAINING DATA for any market figure — CIO standing directive 2026-08-13.

SECTOR RECOMMENDATIONS FOR PATENT AND STEM PARAMETERS

Sector Class	Typical Patent Intensity	Typical STEM Fraction	Examples
EDA Software	0.88–0.92	0.85–0.90	SNPS, CDNS, Siemens EDA — pure IP + software workforce
Semiconductor Equipment	0.85–0.87	0.78–0.83	ASML, AMAT, KLAC, LRCX — hardware but highly IP-intensive
Fabless Semiconductor	0.87–0.90	0.82–0.88	NVDA, AMD, Qualcomm, Broadcom — pure design
IDM (Integrated)	0.80–0.85	0.70–0.78	Intel, Samsung — mixed manufacturing + design workforce
Cloud / AI Platform	0.80–0.88	0.75–0.85	Microsoft, Amazon AWS, Meta — large non-STEM ops workforce
Generic Technology	0.50 (default)	0.50 (default)	Use when sector-specific values unavailable

COMPOSITE SIGNAL CONSTRUCTION

The `composite_signal` is a simple average of all signals that fire: `HURST_MEAN_REVERSION +0.5` (if $H < 0.45$); `GEOMETRIC_CONVERGING +0.6` (spiral-converging); `GEOMETRIC_CHAOS -0.8` (chaotic geometry); `K_ENABLER_STRUCTURAL_PREMIUM +0.9` ($K \geq 0.80$); `K_INTENSIVE_PREMIUM +0.5` ($K \geq 0.60$); `SELDON_MACRO_FAVORABLE +rotation_signal × 0.5` ($rotation > 0.30$); `SELDON_MACRO_UNFAVORABLE` ($rotation < -0.30$); `BACKLOG_CONVERGENCE_STABLE +0.4` ($stability \geq 0.70$). Note: Composite signal is

TACTICAL + STRUCTURAL blended. For long-duration (10-year K-thesis) positioning, weight K-Class and Backlog Stability as primary factors above the composite rank.

PHASE 2D: COMPLETED MILESTONES

Phase	Component	Status	Key Metric
2A	Temporal KAN Models (4 arch × 3 horizons)	☑ COMPLETE	M2 LSTM+KAN avg $R^2=0.562$; M3 MO_KAN $h=1$ $R^2=0.809$
2B	/api/temporal endpoint (FastAPI port 8742)	☑ COMPLETE	blmim_psi_api.py temporal route live
2C	Seldon State Extractor (6-RAG sequential)	☑ COMPLETE	$\Phi(C)=0.3728$; $A=0.2528$ $K=0.84$ $I=0.525$ $\Omega=0.3839$
2D	Seldon Factor Model (7 math frameworks)	☑ COMPLETE	blmim_financial_math.py; EDA sector live analysis

CONVICTION VERDICT: PHI_PLATEAU REGIME — K=1.00 ENABLER STRUCTURAL OVERWEIGHT · SNPS + CDNS LONG-DURATION CORE HOLD

The Seldon Factor Model — applied live against BL3 MySQL price data and EDGAR-cited fundamental ratios — produces the following synthesis for the EDA and semiconductor sector as of 21 August 2026:

PHI(C) MACRO REGIME: PHI_PLATEAU. $\Phi(C) = 0.3728$. The civilisational health variable is stable — neither accelerating improvement nor deterioration. In PHI_PLATEAU, the macro environment is neutral and K-factor selection is the primary alpha source. This is the ideal regime for K=1.00 Enabler structural positioning.

K=1.00 ENABLER TIER: SNPS (\$413.22, PE 91.1×) and CDNS (\$323.47, PE 62.3×). Both are confirmed K=1.00 Enablers (K-scores 0.838 and 0.809). SNPS holds K=1.00 Enabler + Convergence-Stabilized status — the only equity in this universe to achieve both classifications simultaneously. The 1.42× contracted backlog creates a mathematical OU attractor for future revenue (half-life 0.064yr = 16 trading days), giving SNPS the highest structural revenue certainty in the EDA sector.

TACTICAL LAYER: AMAT and LRCX both show spiral-converging phase geometry — they have drawn down and are converging back toward their OU equilibrium levels. This is a tactically tradeable mean-reversion signal, but neither qualifies for structural K-premium (both K-Standard at 0.456 and 0.429). Tactical only, not long-duration.

ASML STRUCTURAL UNDERVALUATION: At PE 54.4× (lowest in the group) with backlog coverage 1.46× (highest) and K-Intensive status (0.655), ASML appears structurally undervalued relative to the K-thesis. The lower K-score reflects hardware R&D constraint (15% R&D/Rev vs EDA SW at 28-31%), but ASML's EUV monopoly confers K=1.00 Enabler status by strategic function even if the scoring formula places it at K-Intensive. The SELDON_FACTOR_LOADINGS assign K=+1.00 to the Semiconductor sector — ASML captures full K-loading despite lower formula score.

BLMIM PHASE 2D VERDICT: The Seldon Factor Model is operational. Seven mathematical frameworks — Hurst (attractor proximity), OU (mean reversion speed), K-Intensity (knowledge capital premium), Seldon Macro Factor (sector rotation), Geometric Phase (trajectory geometry), Backlog Stability (convergence stabilisation), and Phi-Delta Regime (macro classification) — are integrated into a single composite signal backed by BL3 MySQL real price data. Phase 2D integrates all Phase 2 components (2A Temporal KAN + 2B API + 2C Seldon Extractor) into the full BLMIM analytical stack. K = 1.00. The algorithm is running.